

# DINOJAN VIVEKANANTHARASA

dinojv21@gmail.com | +94 779703874 | Vavuniya, Sri Lanka  
linkedin.com/in/dinojan | github.com/Dinojan9901 | Portfolio

## PROFESSIONAL SUMMARY

A fresh Computer Engineering graduate with hands-on experience in **embedded systems, sensor integration, and low-level programming**. Experienced in building microcontroller-based control systems with GSM communication, analog circuit design, and performance-critical C++ development. Combines hardware fundamentals with one year of professional software engineering across systems, simulation, and machine learning.

## EDUCATION

### BSc.Eng (Honours) in Computer Engineering

Mar 2022 – Sep 2026

*Faculty of Engineering, University of Ruhuna, Galle, Sri Lanka*

- **Relevant Coursework:** Embedded Systems, Computer Architecture, Electronics, Signals & Systems
- **Soft Skill Development** (2022–2024)

### GCE Advanced Level

Aug 2018

*V/Vavuniya Tamil Madhya Maha Vidyalayam*

## PROFESSIONAL EXPERIENCE

### Full Stack Developer

Dec 2025 – May 2026

*Yarl Ventures (PVT) Ltd, Jaffna*

- Built EDUS LMS  end to end (Node.js, MongoDB, Redis): RBAC, Socket.IO chat, and PayHere payments.
- Managed MediMan  releases on Linux servers through GitHub Actions CI/CD behind Nginx.

### Full Stack Developer Intern

May 2025 – Nov 2025

*Yarl Ventures (PVT) Ltd, Jaffna*

- Built MediMan admin features with server-side pagination and field projection to cut API payload sizes.
- Integrated Twilio SMS/OTP, Nodemailer, and Firebase push; debugged production via CloudWatch logs.

## TECHNICAL SKILLS

**Embedded & Hardware:** Arduino, C/C++, GSM (AT Commands), UART, I2C, Sensor Interfacing, Relays

**Electronics:** Analog Circuit Design, BJT Amplifiers, Circuit Analysis, Sensor Calibration

**Programming:** C++17, Python, Java, JavaScript, Bash, SQL

**Systems & Tools:** Linux, MPI, OpenMP, CUDA, NS-3, Git, Docker, AWS

## PROJECTS

### GSM-Based Smart Home Automation System

Jun 2024 – Nov 2024

*Arduino, C++, GSM Module, Sensors, Relays*

- Built a microcontroller control system that monitors and switches the home over the cellular network via GSM, so it works without Wi-Fi or internet access.
- Integrated temperature, gas, and LDR sensors with calibration routines, relay switching, and LCD output.

### Parallel Graph Processing (HPC)

Dec 2025 – Apr 2026

*C++17, MPI, OpenMP, CUDA*

- Implemented parallel BFS in OpenMP (atomic CAS, thread-local frontiers), MPI (1D partitioning, Allgather), and CUDA (CSR), all matching a serial baseline exactly.
- Amdahl analysis: 89.4% parallel fraction, 9.43x CPU speedup ceiling; gains emerge above 100k vertices.

### TRACES – Secure Vehicular Network Simulation

Dec 2025 – Sep 2026

*NS-3, Python, Post-Quantum Cryptography*

- Modelled a 200-vehicle wireless network in NS-3 with post-quantum secured communications (ML-KEM-1024, Falcon-1024) and multi-stage anomaly detection.

## Common-Emitter BJT Amplifier

May 2025

*Analog Electronics, Circuit Design*

- Designed and analysed a common-emitter amplifier stage: biasing, operating-point stability, and gain.

## IoT Room Occupancy Detection

Jun 2024 – Nov 2024

*Python, Scikit-learn, Sensor Data*

- Built classifiers on multi-sensor IoT readings for smart-building occupancy detection, end to end.

## Healthcare Mobile Application

Jun 2024 – Nov 2024

*Flutter, Dart, C++, Swift, Firebase*

- Built a cross-platform mobile app with native C++/Swift platform integration, real-time Firebase sync, and authentication.

## PUBLICATIONS

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G. S. J. Venujan, M. F. A. Munsif, **V. Dinojan**, T. Jegan, and G. G. N. Sandamali, “AutoGradePro: An Automated Grading System for Short-Answer Assessments Using Rule-Based Methods and Large Language Models,” in *Proceedings of Academic Sessions 2026*, University of Ruhuna, Sri Lanka, Mar. 2026, Abstract ABS-064. ISSN 2362-0412.

## VOLUNTEER EXPERIENCE

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### Demonstrator – ReXtro 2025 Silver Jubilee Engineering Exhibition

Dec 2025

*Software Engineering Zone, Faculty of Engineering, University of Ruhuna*

### Member

Apr 2024 – Present

*IESL, Galle*

### Member

Sep 2023 – Dec 2024

*IEEE Student Branch, Ruhuna, Galle*

### Treasurer

May 2021 – Apr 2024

*All University Students’ Development Association, Vavuniya*

### Senior Prefect

Jul 2017 – Jul 2018

*V/V.T.M.M.V, Vavuniya*

## ACHIEVEMENTS

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Participated in **IEEE Xtreme 18.0 (2024)**, a global 24-hour algorithmic coding contest (3-member team)  
Competed in **IEEE INSL Southern Provincial Competition**, 30th October 2024

## REFEREES

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- Dr. Kushan Sudheera**

Senior Lecturer

Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna

Contact: +94 719693164 Email: kushan@eie.ruh.ac.lk

- Dr. Nadeesha Sandamali**

Senior Lecturer

Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna

Contact: +94 719693318 Email: nadeesha@eie.ruh.ac.lk